AI Risk Assessment Analysis

Anvay Gaikwad - Sairaj Yadav - Yashashwini Awate - Yogesh Janjire

Otto-von-Guericke University Magdeburg – Faculty of Informatics (FIN)

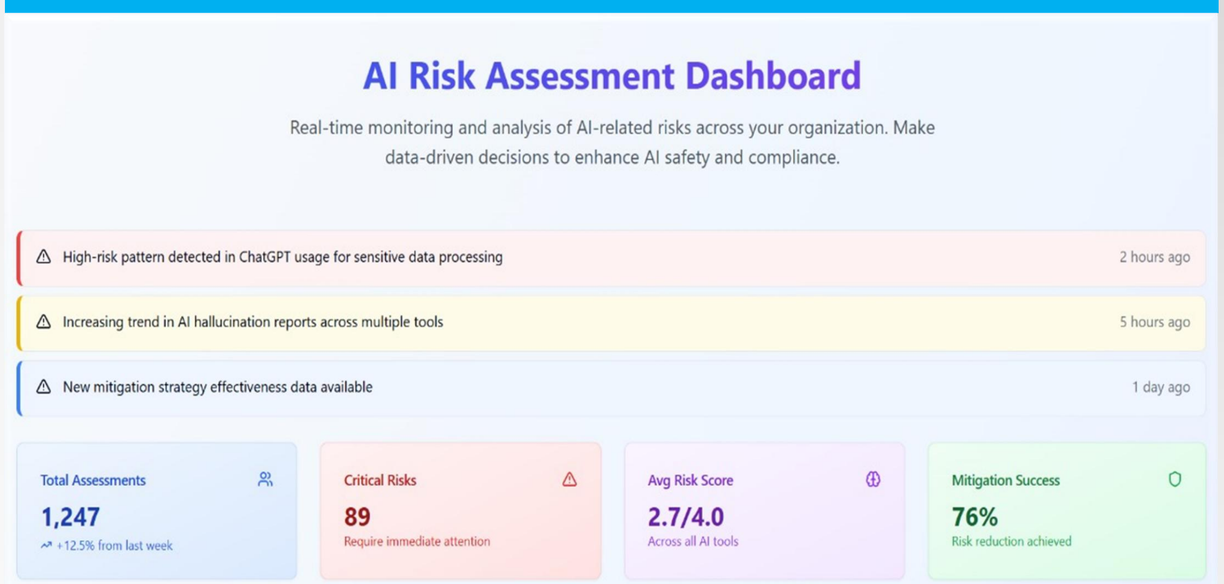
Introduction

This project introduces a comprehensive, data-driven platform for AI risk management, integrating real-time data collection, automated analysis, and an interactive dashboard. Key features include a Google Colab-executable backend for data cleaning, machine learning models with 87.3% accuracy identifying four risk profiles, and over 25 mitigation strategies rated by effectiveness and difficulty.

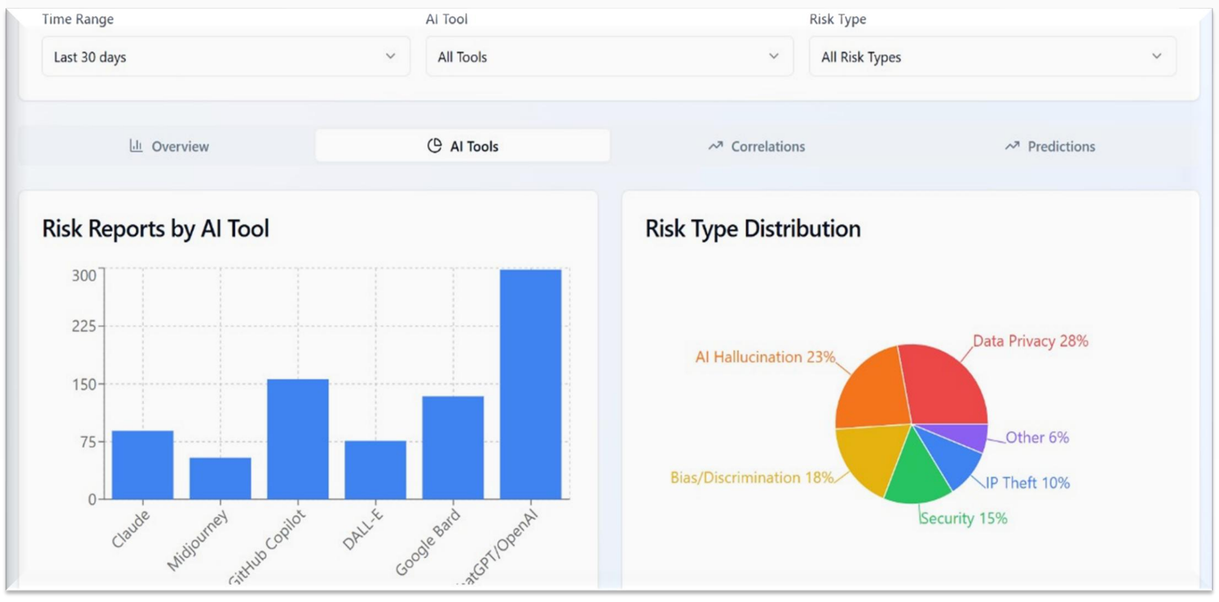
Methodology

A multistage pipeline was developed to assess AI-related risks across education, healthcare, business, and technology. Over 2000 entries from 15 tools were collected, cleaned using imputation and IQR-based outlier removal, and filtered via a quality scoring system. Statistical analyses identified sector-specific risk patterns. A Random Forest classifier estimated risk intensity, while K-Means clustering revealed latent risk structures. Results were visualized through an interactive React/TypeScript dashboard with sector-specific filters and exportable reports

Results



The AI Risk Assessment Dashboard processed 1,247 risk assessments, identifying 89 as "Critical." It flagged high-risk patterns in ChatGPT's handling of sensitive data and noted a rise in AI hallucination incidents. The average risk score was 2.7 out of 4.0, with a mitigation success rate of 76%. These metrics highlight the system's ability to support timely and effective risk management. Tool-specific analysis revealed unique risk patterns, challenging generic assessment methods.



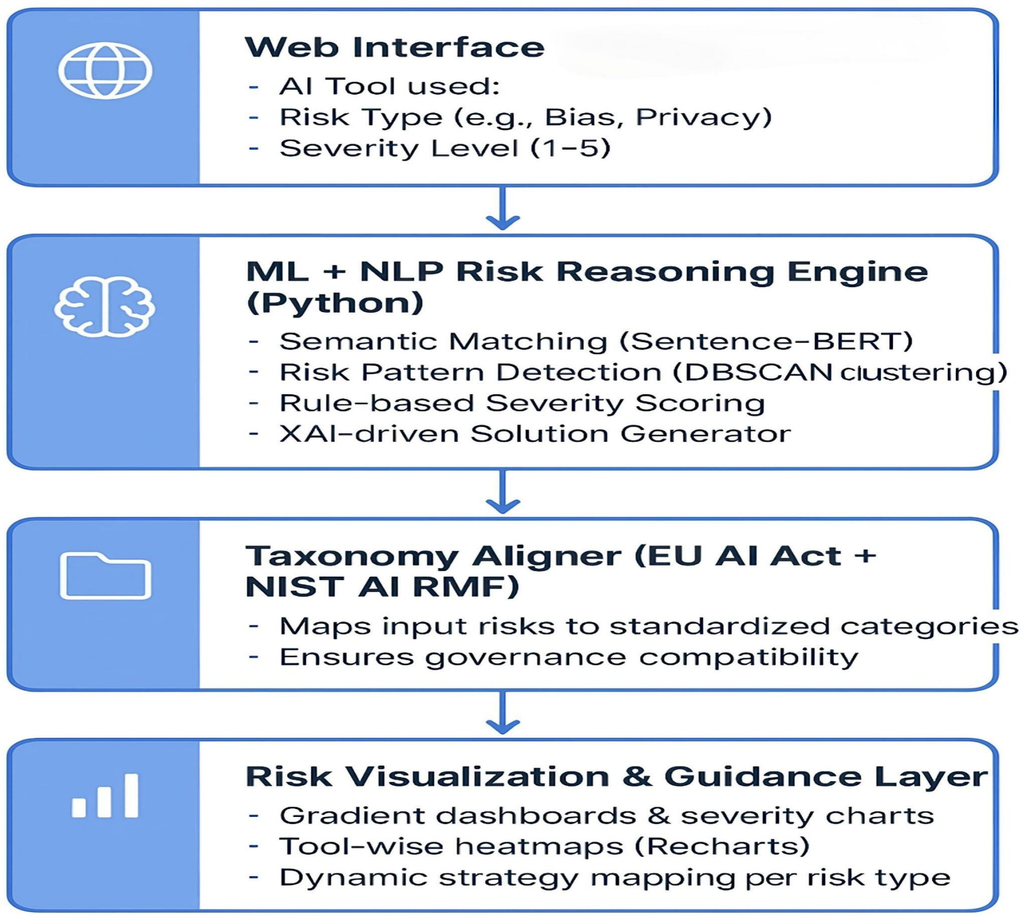
Objectives

The objective of this project is to develop a proactive, data-driven solution that overcomes the limitations of traditional AI risk management. By converting unstructured, subjective inputs into structured intelligence, the system aims to bridge the gap between risk identification and mitigation. It also provides transparent, user-friendly strategies to help organizations effectively manage evolving AI risks.

Literature Review

AI risk management draws on enterprise risk, data science, and HCI. Enterprise platforms (e.g., IBM Watson, Credo AI) offer robust governance. Still, they are large-scale and less accessible, while open-source tools (e.g., Apache Atlas) support metadata and lineage but lack integration and user-focused mitigation. Studies emphasize effective risk visualization using visual hierarchy, color consistency, and interactivity. Recent research also supports ML-based data cleaning methods, highlighting the need for high-quality data in reliable risk analysis.

System Architecture



Data Sources

The dataset was compiled from four key sources: internal controlled studies, peer-reviewed academic research, expert evaluations from real-world AI deployments, and publicly available datasets documenting AI failures and near-miss incidents. These diverse sources ensured comprehensive and reliable risk coverage.

Conclusion

The AI Risk Assessment Analysis System is a scalable, full-stack platform that combines machine learning, data analytics, and interactive dashboards to identify and mitigate AI-related risks across sectors. It supports over 2,000 risk assessments and 25+ mitigation strategies, offering real-time insights for policymakers, developers, and researchers. Future enhancements will focus on expanding use cases, API integration, and usability testing.